Rosseland Mean Opacity for Expansion Opacity from Bound-Bound Lines

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Abstract

We present a new algorithm for calculating Rosseland mean opacity that accounts for expansion opacity effects from bound-bound atomic transitions. The method smooths the frequency-dependent opacity using a $\tau = 1$ window, where for each frequency the distance required to reach optical depth unity is calculated. The smoothed opacity is then determined from $\kappa = 1/(\rho\Delta x)$, where Δx is the path length for $\tau = 1$. Monte Carlo radiative transfer simulations using this new Rosseland opacity yield results similar to full frequency-group calculations, demonstrating that the averaging successfully captures the essential physics of expansion opacity.

Keywords:
opacity, radiative transfer, Rosseland mean, expansion opacity, bound-bound transitions

1. Introduction

The interaction between radiation and matter is primarily determined by the material properties such as composition, density, and temperature, as well as the spectral characteristics of the radiation. In expanding plasmas, due to the Doppler effect, as photons travel through the plasma, their frequency changes. This effect is mainly important for bound-bound transitions, due to their high gradient in frequency which cause photons to interact with different lines as they travel through the plasma. This effect is known as expansion opacity, and it is important for many astrophysical applications, such as Type Ia supernovae and neutron star mergers. Since in most applications, the velocity of the plasma is no more the few percents of the speed of light, the frequency shift due to the Doppler shift is approximately $\Delta\nu = \nu_0 \frac{\Delta v}{c}$.

For non-moving plasmas, the radiation flux can be usually approximated well using the Rosseland mean opacity, which is a weighted average of the opacity over all frequencies. However, for expanding plasmas, the Rosseland mean opacity is not a good approximation, since the opacity changes as the photon travels through the plasma. And therefore the average distance a photon travels depends on his direction of movement and the plasma's velocity gradient.

In this paper we present a new algorithm for calculating Rosseland mean opacity that properly accounts for expansion opacity effects from bound-bound lines. The method assumes a photon which travels "outwards" from the center of the expanding plasma and assumes bolometric expansion ($r = v \cdot t$) and thus gives the opacity as a function of time in addition to the density and temperature $\kappa(T, \rho, t)$.

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The structure of the paper is as follows: In Section 2 we describe how the expansion opacity is calculated, and define an effective expansion opacity that is used to calculate the Rosseland mean opacity. In Section 3, we show the results of the effective Rosseland opacities and discuss their physical meaning. In Section 4 we demonstrate the use of the effective opacity in radiative transfer simulations and compare the results to the full frequency-group calculations. In Section 5 we summarize the results and discuss future work.

2. Expansion Opacity

The total opacity $\kappa_{\rm tot}$ is the sum of contributions from different physical processes:

$$\kappa_{\rm tot} = \kappa_{\rm th} + \kappa_{\rm ff} + \kappa_{\rm bf} + \kappa_{\rm bb} \tag{1}$$

where $\kappa_{\rm th}$ is Thomson scattering, $\kappa_{\rm ff}$ is free-free opacity, $\kappa_{\rm bf}$ is bound-free opacity (which we neglect in this work), and $\kappa_{\rm bb}$ is bound-bound opacity from atomic line transitions.

The BB opacity $\kappa_{\rm bb}$ depends on the plasma's ionization and excitation populations for each material. This is calculated using the Boltzmann distribution.

$$N_{m,i+1} = \frac{N_{m,i}}{N_{m,0}} \frac{Z_{m,i+1}}{Z_{m,i}} \exp\left(-\frac{E_{i+1} - E_i}{kT}\right) \cdot g_e \tag{2}$$

where m, i, j are the indexes of the material, ionization level and energy level respectively, N is the number density, g and E are the degeneracy and energy of the energy level respectively, k is the Boltzmann constant, T is the temperature and g_e is the constant factor $g_e = \left(\frac{2\pi m_e kT}{h^2}\right)^{3/2}$.

$Z_{m,i}$ is the partition function of the state i for the material m, and is calculated using the partition function formula:

$$Z_{m,i} = \sum_j g_j \exp\left(-\frac{E_j}{kT}\right) \tag{3}$$

The number density $N_{m,i}$ is calculated using the Saha equation.

$$\frac{N_{m,i}}{N_{m,0}} = \frac{g_i}{g_0}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} \exp\left(-\frac{E_i}{kT}\right) \tag{4}$$

where m_e is the electron mass.

The optical depth of the bound-bound line is then calculated using the Sobolev approximation.

$$\tau_{j_l,j_u} = t \cdot N_{m,i,j_l} \cdot f_{j_l,j_u} \cdot \frac{\pi e^2}{m_e c} \tag{5}$$

where j_l and j_u are the lower and upper energy levels respectively and f_{j_l,j_u} is the oscillator strength of the transition from the energy level j_l to the energy level j_u, t is the time from explosion.

Then, the opacity is calculated as:

$$\kappa_{BB}(\nu) = \sum_l \delta(\nu - \nu_l) \cdot \tau_l \tag{6}$$

For a steady plasma, the Rosseland mean opacity gives the average distance energy travels before being scattered or absorbed. The Rosseland mean opacity is calculated as:

$$\frac{1}{\kappa_{\rm R}} = \frac{\int_0^\infty \kappa_\nu^- 1 \frac{\partial B_\nu}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu}{\partial T} d\nu} \tag{7}$$

where B_ν is the Planck function and T is the temperature.

The naive application of Rosseland mean opacity to expansion opacity does not work for medium with high BB contribution because BB lines have no width and therefore their contribution is zero.

Therefore, up to now, the radiation transport for expanding systems with significant BB opacity was calculated by tracking photons with Monte Carlo simulations. The photon's frequency was changed according to the Doppler shift, and each time a photon reached a new spectral line, the photon was absorbed according to the probability for absorption which is $P = 1 - \exp(-\tau_{j_l,j_u})$. A outward travelling photon with frequency ν_0, will reach a new spectral line with frequency ν_l after a velocity shift of $\Delta v = -c\ln(\nu_0/\nu_l)$ which is corresponding to a distance of $\Delta x = -ct\ln(\nu_0/\nu_l)$.

For the purpose of calculating effective Rosseland opacity, we calculated the average distance (and equivalently frequency shift) an outward travelling photon goes through in a plasma with density ρ, temperature T and time t will travel before interacting with the plasma. For that we define the frequency optical depth as:

$$\tau(\nu) = \int_{\nu_{\rm min}}^{\nu_{\rm max}} \kappa_\nu \rho \, dx = ct \int_{\nu_{\rm min}}^{\nu_{\rm max}} \kappa_\nu \rho \frac{d\nu}{\nu}. \tag{8}$$

Then, the typical frequency shift $\Delta\nu$ a photon with frequency ν_0 goes through before interacting with the plasma fulfills: $\tau(\nu_0 + \Delta\nu) - \tau(\nu_0) = 1$. Therefore, we can define a new expansion opacity, $\kappa_{eff}(\nu) = \frac{1}{\rho\Delta R}$:

$$\kappa_{eff}(\nu) = \frac{1}{\rho ct \frac{\Delta\nu}{\nu}} \tag{9}$$

For numerical purposes, one can discretize the frequency and calculate an opacity for each bin $\tau(\nu_i)$. In that case one needs to handle the case of a bin with optical depth jump of more than 1. In that case, using $\Delta\nu = \nu_{i+1} - \nu_i$ is not optimal and creates an artificial opacity maximum of $\frac{1}{\rho ct \frac{\nu_{i+1}-\nu_i}{\nu_i}}$. This maximum opacity might be correct for a single line, but it is incorrect for multiple lines in the same bin. To avoid that, we calculate those cases separately by:

$$\kappa_{eff}(\nu_i) = \frac{1}{\rho ct \frac{\nu_{i+1}-\nu_i}{\nu_i} \sum_j (1 - \exp(-\tau(\nu_j)))} + \kappa_{th} + \kappa_{ff}(\nu_i) \tag{10}$$

where the sum is over all lines in the bin.

A visual example of the opacity of iron, after 10 days, in a density of $3 \cdot 10^{-12}$ g/cm^3 and a temperature of 1.5 eV, is shown in Figure 1. The opacity is shown once for the raw opacity with 1M bins (in logarithmic resolution) , then for the effective opacity, once with 1M bins and once with 300 bins. In the case of 300 bins, the BB contribution is still calculated through the formalism in Eq. (10). The opacity calculation used the atomic line data of Kurucz & Bell (1995; CD 23) [1].

3. Effective Rosseland Mean Opacity Results

Figure 2 shows the convergence of the Rosseland mean opacity as a function of the frequency resolution $\Delta\nu/\nu$ (which is equivalent to the number of frequency bins in logarithmic resolution). We compare three different approaches: the naive opacity (calculated without smoothing), the naive saturated opacity, and the effective opacity (calculated using the expansion opacity formalism with $\tau = 1$ window smoothing). Both naive opacities use Eq. (10) for each bin separately, but differ in whether they include the exponential term: the naive unsaturated opacity uses $\tau(\nu_j)$ directly (without the exponential), while the naive saturated opacity uses $(1 - \exp(-\tau(\nu_j)))$ (with the exponential).

The results show that the effective opacity is barely dependent on the frequency resolution and gives converged Rosseland opacity even for low resolutions (large bins) such as $\Delta\nu/\nu \gtrsim 4 \cdot 10^{-2}$ (which is equivalent to 100 bins). The reason the effective opacity gives correct results even for large bins is that the spectral lines are randomly distributed within each frequency bin. As long as the lines are randomly distributed, Eq. (10) provides a good approximation since it represents the typical distance a photon travels in that frequency range. The fact that we observe good agreement between low-resolution and high-resolution calculations for the effective Rosseland mean confirms that the line locations are indeed close enough to randomly distributed within the bins. The naive saturated opacity

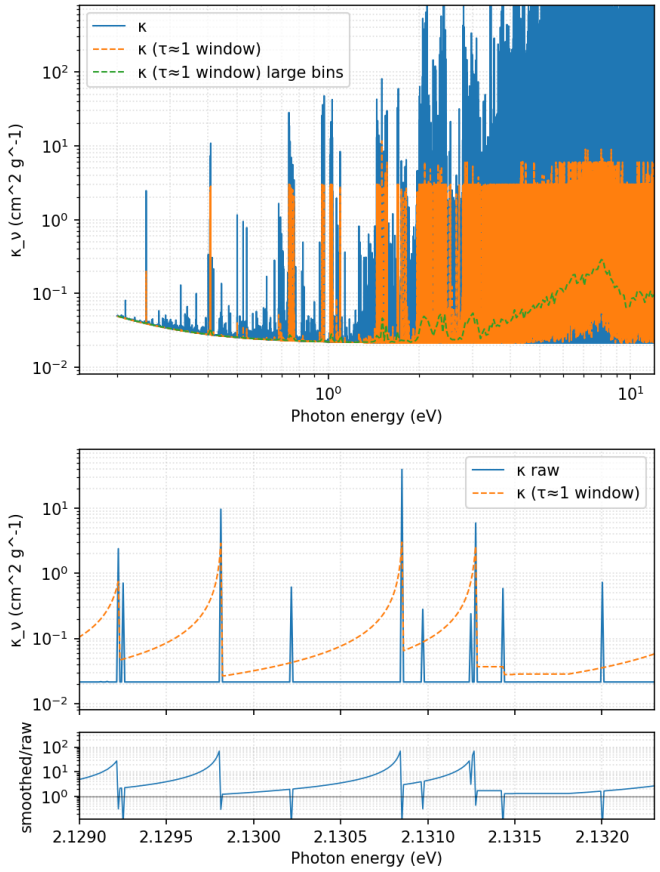


Figure 1: Examples of bound-bound line opacity showing the $\tau = 1$ window smoothing algorithm. Left: illustration of the smoothing process. Right: comparison of raw opacity (solid line), smoothed opacity with fine frequency bins (dashed line), and smoothed opacity with large frequency bins (dotted line), demonstrating the effect of the $\tau = 1$ window smoothing on bound-bound line transitions.

(using the expansion opacity formula but without $\tau = 1$ window smoothing) works reasonably well at low resolution because it is actually equivalent to the effective opacity in this regime. This equivalence occurs because at low resolution, each frequency bin has $\tau > 1$, and in this case the naive saturated opacity and the effective opacity are identical. For smaller frequency bins ($\Delta\nu/\nu \lesssim 10^{-4}$) the lines are so narrow they cannot affect the Rosseland opacity and the naive saturated opacity gives zero opacity. The naive opacity (without smoothing) gives high opacity for large bins where the lines are completely smoothed out but still maintain their volume. For small bins (high resolution) we get almost identical result to the saturated opacity since the opacity addition of a bin in Rosseland opacity goes as $\frac{1}{\kappa_\nu}$. Therefore Rosseland opacity is not sensitive to the extent in which lines increase a single point as long as they increased it by more than the continuum opacity.

Using the formalism formerly described, we calculated the opacity of materials for different times, densities and temperatures and constructed "libraries" that can be used for simulations. For materials which decay with time such as Nickel 56, we also calculated their decay rate and for each time window used the composition which fits that time. We assumed

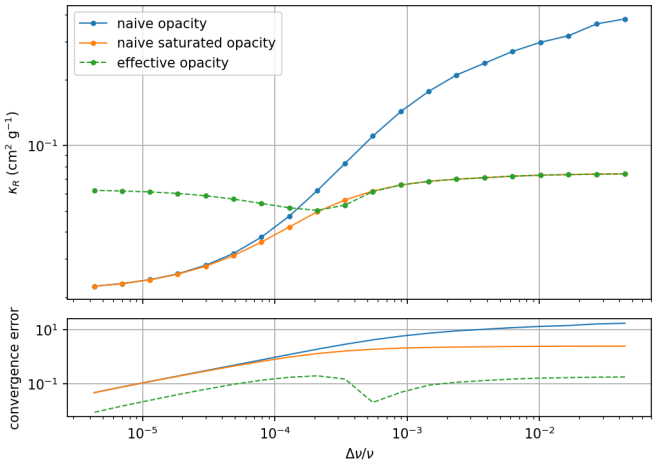


Figure 2: Convergence of Rosseland mean opacity as a function of the frequency resolution $\Delta\nu/\nu$. Three approaches are compared: naive opacity (without smoothing), naive saturated opacity (with expansion opacity formula but without $\tau = 1$ window smoothing), and effective opacity (with full expansion opacity formalism including $\tau = 1$ window smoothing).

the decay rates of $t_{1/2}(^{56}Ni) = 6.075$ days and $t_{1/2}(^{56}Co) = 77.233$ days.

Figures 3 and 4 show the effective Rosseland mean opacity for nickel as a function of temperature and density (Figure 3) and as a function of temperature and the product $\rho \cdot t$ (Figure 4) at three different times (1, 10, and 100 days). The opacity increases with temperature and decreases with density and time, as expected for bound-bound transitions in expanding media. The $\rho \cdot t$ representation is more natural for expanding media and better illustrates how the opacity scales with the expansion parameter.

The opacity behavior shown in figures 3 and 4 can be understood from the expansion opacity formalism. From Eq. (9), the effective opacity is $\kappa = 1/(ct\rho\Delta\nu/\nu)$. For bound-bound dominated frequencies, photon absorption is mainly limited by saturated lines, which will absorb the photon regardless of the density or time since $\tau \gg 1$. In this regime, $\Delta\nu/\nu$ is approximately constant, and therefore $\kappa \sim 1/(\rho \cdot t)$.

However, the opacity shown in the figures deviates significantly from this simple scaling for several reasons. Firstly, the bound-bound contribution is only important for certain frequency ranges, while at other frequencies free-free and Thomson scattering dominate the opacity. Secondly, the Saha equation solution depends on density, and therefore the ionization fraction changes with density, affecting the bound-bound line strengths. Thirdly, the time evolution changes the composition (^{56}Ni to ^{56}Co to ^{56}Fe), which has slightly different opacities. These effects combine to produce the complex behavior observed in the opacity maps. Still, the $\rho \cdot t$ representation in Figure 4 shows less time sensitivity than Figure 3 because the time dependence is largely absorbed into the scaling parameter, making the opacity behavior more uniform across different times.

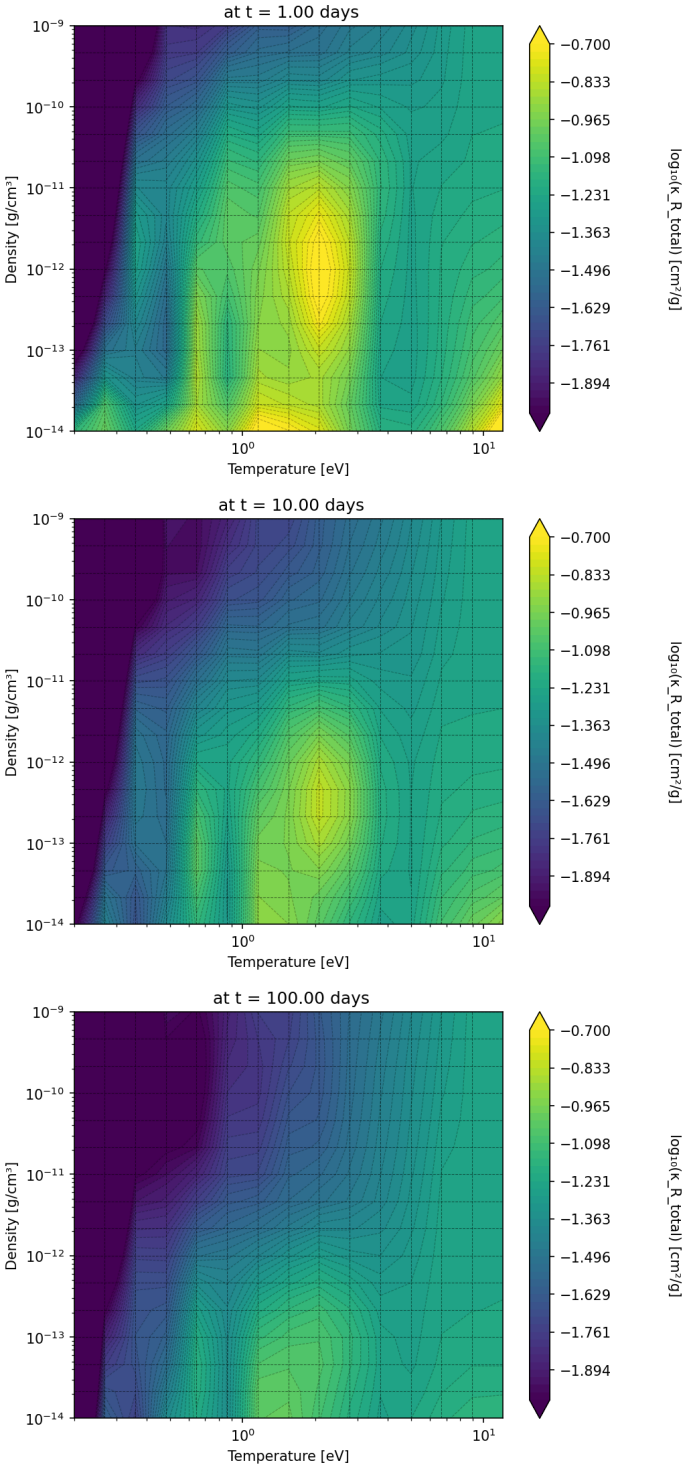


Figure 3: Effective Rosseland mean opacity for nickel as a function of temperature and density at t = 1, 10, and 100 days (from left to right). The opacity is calculated using the expansion opacity formalism described in Section 2.

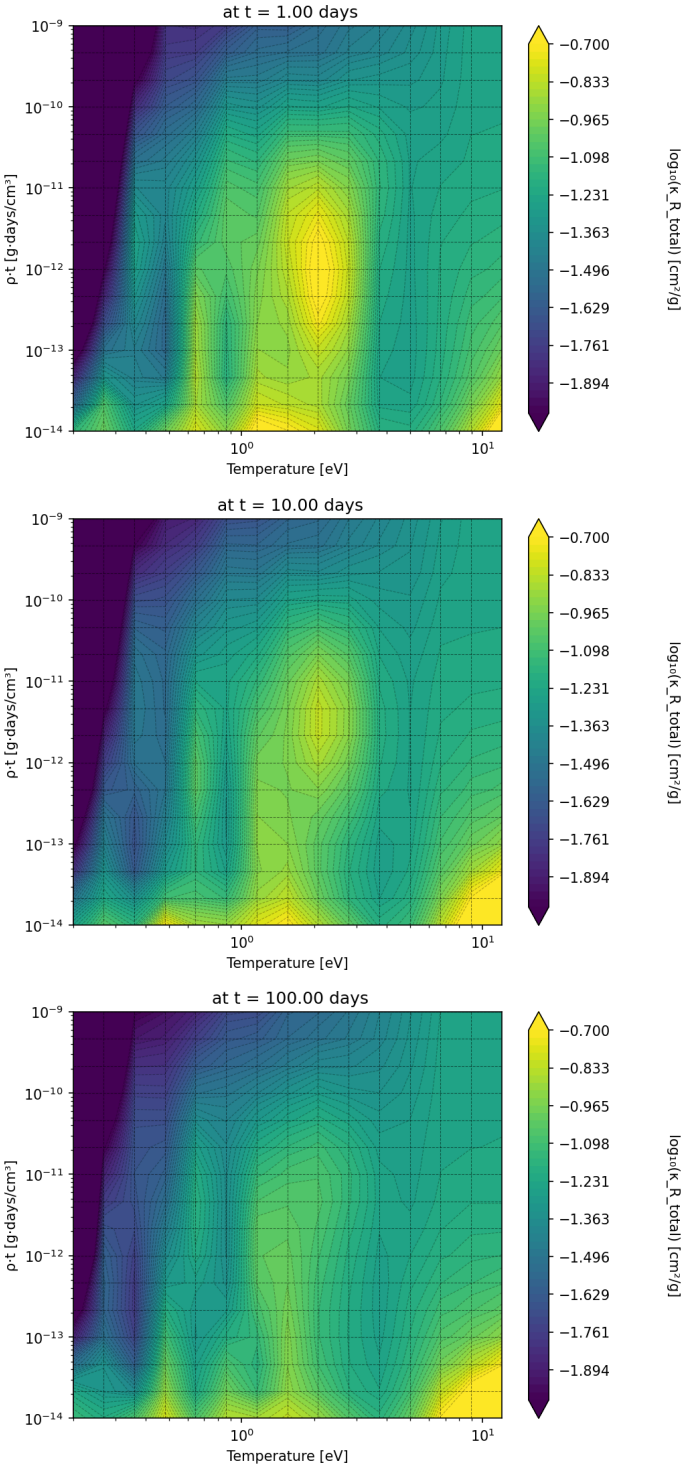


Figure 4: Effective Rosseland mean opacity for nickel as a function of temperature and $\rho \cdot t$ at t = 1, 10, and 100 days (from left to right). The $\rho \cdot t$ parameter is more natural for expanding media and better illustrates the scaling behavior of expansion opacity.

4. Validation

Monte Carlo radiative transfer simulations using the new Rosseland mean opacity calculated from the $\tau = 1$ window algorithm yield results similar to full frequency-group calculations. The simulations were performed using URILIGHT [2, 3], an open-source Monte Carlo radiative transfer code designed for supernova ejecta. URILIGHT solves the time-dependent radiative transfer equation in expanding media using a Monte Carlo approach, tracking photon packets through the ejecta and accounting for bound-bound, free-free and thomson opacity sources.

In Figure 5 we compare the results of the new Rosseland mean opacity to the full frequency-group calculations. We performed two types of simulations. The first uses 100 cells with an exponential decay density profile, representing a more realistic supernova structure. This simulation shows reasonable but not perfect agreement (approximately 25% deviation) with the full frequency-group calculations. The discrepancy arises because as the temperature decreases from the center of the supernova toward the outer regions, photons emitted in the center have a higher probability of being absorbed, since the opacity to high frequencies is generally higher. This effect is not captured in the gray simulation because the gray formalism does not account for temperature gradients.

To test this hypothesis, we performed a second simulation using a single cell with a single temperature. For this case, the gray model shows much better agreement with the full frequency-group calculations, confirming that the gray formalism works well when temperature gradients are absent. This demonstrates that the averaging process successfully captures the essential physics of expansion opacity, at least for the case of constant temperature.

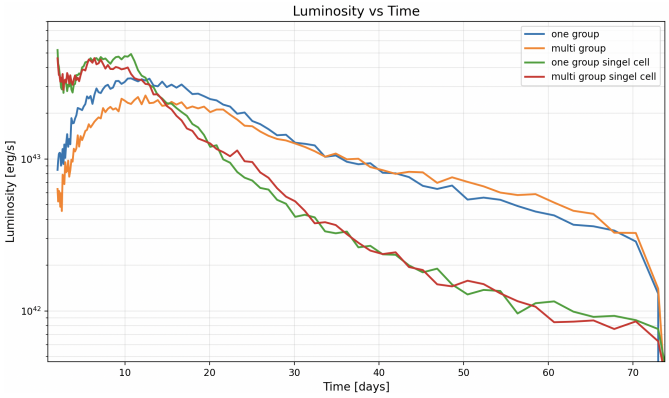


Figure 5: Comparison of Monte Carlo radiative transfer results using the new Rosseland mean opacity (from $\tau = 1$ window algorithm) versus full frequency-group calculations. Left: 100-cell simulation with exponential density profile showing approximately 25% deviation. Right: single-cell simulation with uniform temperature showing excellent agreement. The discrepancy in the multi-cell case arises from temperature gradients, which the gray formalism does not account for.

5. Summary

We have presented a new algorithm for calculating Rosseland mean opacity that accounts for expansion opacity effects from bound-bound atomic transitions. The method uses a $\tau = 1$ window to smooth the frequency-dependent opacity, with the smoothed value calculated as $\kappa = 1/(\rho \Delta x)$ where Δx is the path length to reach optical depth unity. Validation through Monte Carlo simulations confirms that this approach captures the essential physics while enabling efficient gray opacity calculations.

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